Ching-Hsiang Wu

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EDUCATION

National Taiwan University Taipei, Taiwan
Master of Science, Automatic Control in Electrical Engineering Feb. 2025
Bachelor of Science, Biomechatronics Engineering Feb. 2022
Relevant Courses: Automatic Control, Digital Control, Adaptive Control, Optimal Control, Reinforcement Learning, Nonlinear system analysis, 2D/3D Computer Vision.

SKILLS

Programming: Python, C/C++, MATLAB, Qt.
Robotics: ROS/ROS2, Isaac Sim/Lab, Rviz, Gazebo, SolidWorks, Simulink.
Controller: Raspberry pi, Arduino, Nvidia TX2/Xavier, Pixhawk series, PX4.
Artificial Intelligent: Deep learning, Reinforcement learning.
Specialty: Robot manipulator kinematics/dynamics, Control system, Computer vision.

RESEARCH EXPERIENCE

Inventec Inc. Taipei, Taiwan
Robotics engineer Nov. 2025-Now
Moveit2 and ROS2 on an autonomous robotic system
- Validate the function of the robotic system through RViz and Isaac Sim simulator.
- Study admittance and impedance control to ensure robot safety.
- Sensorless external wrench estimation implementation in Isaac sim.

Networked Control System Laboratory (NCSLab) Taipei, Taiwan
Graduate student Feb. 2023-Feb. 2025
Fixed-wing UAVs formation flight under variant wind disturbances
- Design the Lyapunov-based formation controller and sliding mode wind observer for fixed-wing UAVs.
- Validate the formation flight performance via SITL simulation, integrated with PX4, Gazebo, and ROS2.

AiSeed Tech Inc. Taipei, Taiwan
Robotics AI engineer intern Oct. 2021-Aug. 2022
Validate VTOL UAV systems functionality with ROS and Gazebo simulator
- Design a landing algorithm with a changeable landing position for VTOL UAVs.
- Stream inferred video from UAV system to website or ground station through Gstreamer.

Robots and Medical Mechatronics Laboratory (RMML) Taipei, Taiwan
Undergraduate researcher Sept. 2019-Sept. 2021
Develop a platform for remote control robots for oral and nasal cavity specimen collection
- Build an autonomous specimen collection robot with remote center motion (RCM) mechanism.
- Win sponsorship from the Ministry of Science and Technology (MOST) for 800000 NTD dollars.

LEADERSHIP EXPERIENCE

RL Final Project – Q-Drive Taipei, Taiwan
Use PPO to train a quadruped to reach a desired position with tripod gait Oct. 2024-Dec. 2024
- Leverage curriculum learning and hierarchical RL techniques to utilize both front legs of quadrupeds.
- Effectively divide work to team member and organize the weekly meeting to sync up the project progress.
- Win the first 10th place out of 25 groups in the final presentation competition.

Azalea Festival Project Taipei, Taiwan
Build an autonomous sensing and catching apple car system Feb. 2020-Mar. 2020
- Grab apples with a RRR manipulator automatically by obtaining apples' 3-D coordinates.
- Train a model to recognize green and red apples with tiny-yolov3.

PUBLICATION

Lyapunov-Based Formation Controller Design for Fixed-Wing UAVs under Variant Wind Fields
- Wu and Lian, International Conference on Advanced Intelligent Mechatronics (AIM2025)